



PAPER

Numerical investigation of highly viscous droplet generation based on level set method

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E-mail: gzzysy_zc@163.com, tomn_in@163.com and ling_mx@163.com**Keywords:** high-viscosity, drop-on-demand, droplet formation, level set method, inkjet printing**Abstract**

Piezo-driven needle valves are widely used in electronic packaging due to their fast response, high resolution and good dispensing consistency. However, the stable generation of high-viscosity droplets is one of the key issues to its packaging quality. To investigate the formation mechanism of the high-viscosity droplet. In this paper, a 2D finite element model of the drop-on-demand injection process of the high-viscosity droplet is established based on the level set method, the droplet formation and separation processes are numerically simulated, and the reliability of the simulation results is checked by comparing the outcomes with published data. Specifically, the detailed evolution of the high-viscosity droplet formation and separation process is gained by coupling the two-phase flow-level set method and the dynamic grid technique, and the pressure distribution in the injection chamber is further discussed and the effects of operating parameters on the droplet formation volume are examined. The results of the study show that the needle motion is the main factor of pressure fluctuations in the injection chamber. Moreover, we also found that among the parameters of needle stroke, nozzle diameter, supply pressure, fluid viscosity, and surface tension, the nozzle diameter has the most significant effect on droplet volume, while surface tension has the least effect on droplet formation.

1. Introduction

Micro-droplet jetting technology is an exactly controlled manner of distributing adhesive materials to the surface of electronic circuit components and chips to accomplish the connection and protection between microelectronic components. Over the last few years, benefiting from the continuous enhancement of jet performance in terms of precision, efficiency, and repeatability of Micro-drop jetting technology, its applications are no longer limited to traditional inkjet printers, but have displayed tremendous application foreground in several emerging areas such as micro-electronic packaging [1, 2], micro-nano manufacturing [3, 4], and biomedicine [5, 6].

Among multitudinous application scenarios, the jetting process of highly viscous fluids has always been an interesting and complex issue. To realize the injection of high-viscosity fluids, a common method of dispensing high viscosity fluid materials is to utilize a needle-type dispenser. In this method, a localized high pressure is created in the working chamber by manipulating the movement of the needle, thereby forcing the adhesive to flow out of the nozzle in the form of liquid droplets. Currently, scholars [7–9] have developed needle-type dispensers based on various driving means, which mainly include piezoelectric ceramics, high-pressure gas, or giant magnetostrictive material driving. Among them, piezoelectric ceramic drives have captured extensive attention from scientists as a result of its excellent strength of rapid response, fine resolution, and high output force. For instance, Zhou *et al* [10, 11] proposed a novel type of ejector using piezoelectric stacks as power sources, which can inject fluid with a maximum viscosity of 1.4 Pa·s without heating the liquid. Lu *et al* [12] designed and fabricated a piezoelectric-actuated jetting dispenser with a viscosity of 3 Pa·s employing the

lumped parameter method. Nguyen *et al* [13, 14] developed a piezostack-driven needle collision jetting based on hydraulic or lever principle amplification, which can jet liquid with a viscosity of up to 10 Pa·s. Nevertheless, the foregoing investigations principally lay particular emphasis on the operational performance of the jet dispenser, while the intrinsic mechanism of high-viscosity droplet generation is not studied in depth.

As is well known, the micro-scale droplet formation process is a typical two-phase flow problem, which essentially tracks the topological change at the interface of two immiscible fluids (e.g., the polymer on one side, and the air phase on the other side in this text). In earlier studies, academics [15–17] utilized experimental methodologies to investigate the droplet generation processes and obtained a definite achievement. However, owing to the experimental approach makes use of the image processing techniques to analyze the transient shape of a large number of droplets [18], its modeling accuracy strongly depends on the resolution of an expensive ultra-high-speed camera [19]. Moreover, since the micro-drop jetting dispenser driven by piezoelectric stacks is ordinarily required to generate droplets with sub-millimeter or even micrometer diameters within a few microseconds, it is difficult and laborious to observe droplet formation through experiments [20]. Accordingly, the limitations of experimental research are extraordinarily prominent both in terms of precision requirements and cost management. Fortunately, along with the explosive growth of computing power [21], numerical simulation, with its high efficiency and low cost, has gradually become an alternative solution to handle the two-phase flow. In practice, nevertheless, the micro-droplet formation process is influenced by the computational accuracy of curvature at the free interface, surface tension effects, and drastic changes in the two-phase flow field. Hence, the employment of numerical simulation of the droplet generation process remains a complicated and challenging task.

To date, two well-known numerical algorithms have been devoted to simulating drop formation: the volume of fluid (VOF) method and the level set method (LSM). These two methods have their advantages and shortcoming. For instance, the VOF method has the merit of automatically handling the topological transformation of an interface with surface tension and ensuring the complete conservation of fluid mass. For all that, the color function used in the VOF method is discontinuous, which induces a loss of accuracy in determining the position of the surface with surface tension [22]; usually, only first-order accuracy can be achieved. In addition, a cumbersome interface reconstruction algorithm is required to be introduced to compute the volume of the fraction convection [23]. Despite its many shortcomings, the VOF method is still the dominant numerical method for simulating high-viscosity droplet generation. For example, Lu *et al* [24, 25] numerically investigated the high-viscosity droplet formation and separation process in a needle-type dispenser employing the VOF method. The principal conclusions were obtained that larger supply pressure is favorable for droplet formation, while a larger diameter tends to cause unstable injection. Li *et al* [20] established a CFD model based on the VOF method for the high-viscosity paste injection process and explained the causes of droplet shape defects during droplet formation by combining numerical studies and experiments. Zhou *et al* [26] analyzed the drop-on-demand dispensing mechanism of a needle injector by the VOF method and gained that the needle stroke was the main factor affecting the droplet formation volume. In addition, the VOF method has played an important contribution in the study of mixing performance of high-viscosity fluids, for example, the literature [27] numerically investigated the mixing of high-viscosity fluids in a two-dimensional T-junction under steady-state conditions based on the finite volume method and explored the effects of slug length, bubble and Reynolds number on the performance of the micro-mixer. Compared with the former, the LSM is more advantageous in tracking the topological changes of the free interface. This is because the level set function used to describe the interface change is continuous. This function has two advantages: On the one hand, it can precisely trace the motion of the interface even when it undergoes extreme topological changes, e.g., merging or pinching off. On the other hand, it can still acquire high-order accuracy in calculating the curvature of the interface when the volume fraction of the two-phase fluid near the interface changes drastically in the method. With these many benefits, the LSM has been welcomed among many researchers in simulating droplet generation. For instance, the droplet generation process in microfluidic two-phase flow based on the LSM was explored in [28, 29]. The LSM was conducted to simulate the droplet formation of a piezoelectric jet dispenser, see [30]. In [31], the droplet formation process in an EHD printer under a pulsed electrical field was simulated using the LSM. However, the studies above are primarily focused on low-viscosity droplet formation.

In summary, a considerable number of publications have reported work related to high-viscosity droplet formation based on the VOF method; however, their work does not provide an in-depth discussion of the pressure variation within the injection chamber during high-viscosity droplet generation. In addition, thanks to the success of the LSM in simulating droplet formation from low-viscosity fluids, the use of the LSM to simulate high-viscosity droplet formation is worth further exploration. Based on this, a two-phase flow numerical mode of the needle-type injector is established, and the motion characteristics of the needle are simulated by using the dynamic grid technique based an Arbitrary Lagrangian-Eulerian method. Then, the kinetic behavior of the droplet generation process is obtained by coupling calculations of the two-phase flow-LSM and the dynamic grid, and the causes of pressure fluctuations in the injection chamber during the whole droplet formation

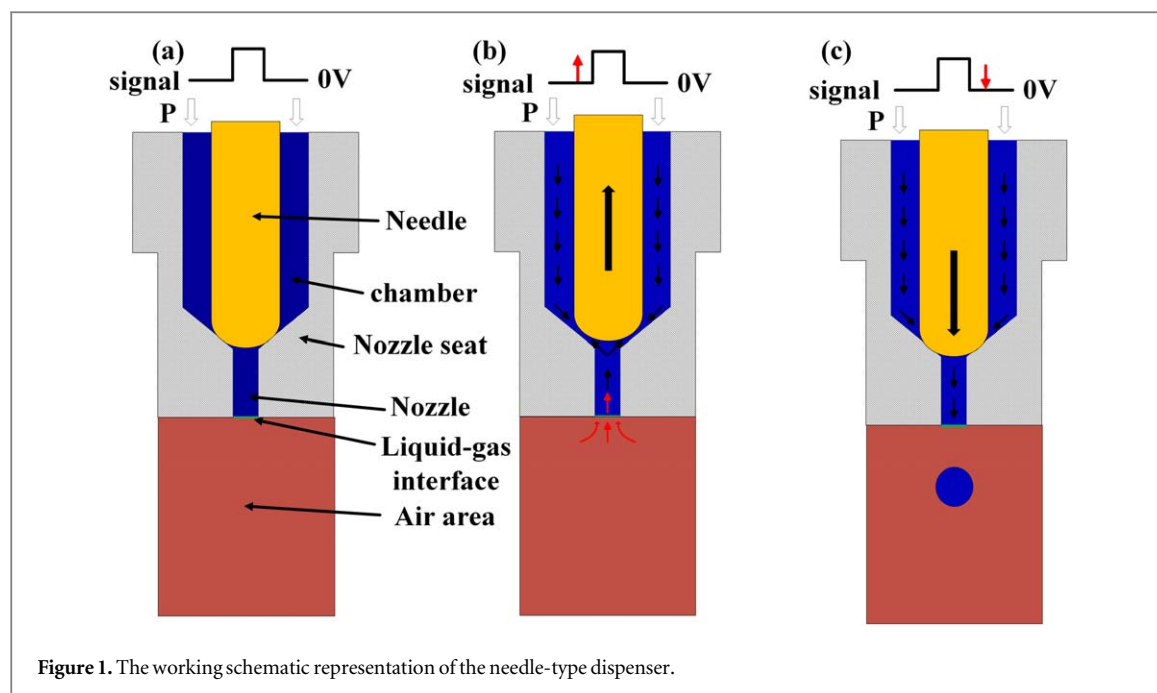


Figure 1. The working schematic representation of the needle-type dispenser.

process are further concretely discussed. Lastly, the relationship between droplet volume and needle stroke, nozzle diameter, supply pressure, fluid viscosity, and surface tension during droplet generation is also examined. The results of this research may offer some guidance for the stable generation of high-viscosity droplets.

2. Mathematical method

2.1. Description of the working process

The working schematic representation of the needle-type dispenser was briefly illustrated in figure 1, which is mostly composed of three sections: the needle part, the work chamber, and the nozzle. The concrete working procedure can be displayed as follows: at the initial moment, the working chamber is filled with liquid, and the needle is in intimate contact with the nozzle. The needle is then driven upwards by the piezoelectric actuators, and outside air enters the nozzle. As the needle moves to the peak point, it stays there for a while, during which the feed fluid will completely occupy the gap created by the needle movement. Next, the needle will move downward, and as the distance between the needle and the nozzle decreases to zero, a local instantaneous high pressure will form inside the nozzle. At the current moment, the kinetic energy gained by the liquid inside the nozzle is sufficient to break through its own resistance and form a jet. In the end, under the joint competition of inertia force, gravity, viscous force, surface tension, and atmospheric pressure, the liquid will break off to form droplets moving toward the substrate.

2.2. Numerical method description

2.2.1. The level set method

The LSM is just an Eulerian approach that can track the topological changes of free interfaces in a two-phase flow on fixed grids. It was originally given by Osher *et al* [32] in the 1980s. The basic principle is to define the level set function as a signed distance function to the interface and implicitly represent the interface location with the zero contour of the level set function. It has shown tremendous application foreground in many areas, including but not restricted to fluid mechanics, flame propagation, computer vision, and material science. However, the classical LSM suffers from a fatal flaw, i.e., it is prone to fluid mass loss. Therefore, academics have propounded a large number of schemes to address the shortcomings of the LSM. The more common ones are the construction of hybrid LSM (as noted in [33, 34]), reducing the spatial dispersion error method (see details in [35]), and designing a LSM with a reinitialization procedure (see [21, 23] for details of the reinitialization process). In this investigation, we choose a LSM with a reinitialization procedure (hereinafter referred to as R-LSM), which has good mass conservation and high-order accuracy. The core idea is to use the contour of the level set function $\phi = 0.5$ defined as the location of the fluid-air interface. When the level-set function $\phi = 0$ or 1 is expressed in the air and fluid regions, respectively. And in the transition layer near the interface, the level set function ϕ varies smoothly from 0 to 1 at the free interface.

2.2.2. Basic governing equation

In this text, we assume that the fluid is an incompressible, low Mach number flow Newtonian fluid. Likewise, the temperature of the liquid remains constant throughout the injection process, i.e., the viscosity and surface tension of the fluid is a fixed value. Else, the effect of gravity on the fluid during the injection procedure is negligible [36]. Consequently, in accordance with the theory of fluid mechanics, the Navier–Stokes equations for the dynamic behavior of an incompressible Newtonian fluid dynamics with the consideration of the surface tension effect can be written in the following forms.

$$\rho \frac{\partial \mathbf{u}}{\partial t} + \rho \mathbf{u} \cdot \nabla \mathbf{u} + \nabla p - \nabla \cdot (\mu \nabla \mathbf{u}) = F_{st} \quad (1)$$

$$\nabla \cdot \mathbf{u} = 0 \quad (2)$$

where ρ , p and $\mathbf{u}$ are fluid density (kg/m^3), pressure (Pa) and velocity vector (m/s), respectively. μ is the dynamic viscosity (Pa·s), ∇ is the gradient operator, F_{st} is defined as the surface tension (N) acting on the fluid interface.

With the purpose of determining the shape and location of the two-phase interface, we employ a R-LSM to trace the free interface. Whose convection transport equation can be expressed as

$$\frac{\partial \phi}{\partial t} + \nabla \phi \cdot \mathbf{u} = \gamma \nabla \cdot [\varepsilon \cdot \nabla \phi - (\phi(1 - \phi)\mathbf{n})] \quad (3)$$

The left-hand term of the above equation is the governing equation of the classical LSM, which represents the direction of motion of the interface, where the value of the velocity field $\mathbf{u}$ is obtained by solving equations (1) and (2). The contribution of the right-hand term of the equation is to ensure that the fluid mass remains conserved in the numerical calculation. Here, γ and ε are numerical stability parameters that denote the number of re-initializations in the two-phase flow calculation and the thickness of the transition layer at the free interface, respectively. The magnitude of the values depends on the specific model, generally γ is the maximum amplitude occurring in the velocity field of the model, and ε is half of the typical grid size of the fluid flow region. $\mathbf{n}$ is the interface normal, which can be obtained by the following equation.

$$\mathbf{n} = \nabla \phi / |\nabla \phi| \quad (4)$$

Additionally, for the purpose of reducing the numerical oscillations that occur when solving the Navier–Stokes equation due to the violent changes in the physical parameters of the fluid at the free interface, we play the level set function ϕ to smooth the liquid density and viscosity at the transition layer of the interface.

$$\rho = \rho_1 + (\rho_2 - \rho_1)\phi \quad (5)$$

$$\mu = \mu_1 + (\mu_2 - \mu_1)\phi \quad (6)$$

among them, ρ_1 , μ_1 , ρ_2 , μ_2 denote the constant density and viscosity of fluid and air, respectively.

Surface tension plays a critical role in the droplet formation [30]. In these previous studies, the surface tension was straightly imposed on the free interface as a boundary value condition [37]. Unfortunately, this modeling mode may induce surface tension distortion when dealing with complex topological changes at the free interface. To enhance the robustness of the numerical computation, here we adopt the Continuous Surface Force model developed by Brackbill *et al* [38] based on the continuum theory. Which introduces the surface tension into the body force source term of the Navier–Stokes equation, and the surface tension acting at the interface is calculated by the following formula.

$$F_{st} = \sigma k \delta \mathbf{n} \quad (7)$$

which σ is the surface tension coefficient (N/m), k is the curvature, and its physical meaning is expressed as the second derivative of the level set function. δ is the Dirac function of the interface, which represents the position of the singular surface tension [39]. Furthermore, to simplify the calculation of surface tension, the δ function can also be approximated by equation (8).

$$\delta = 6|\nabla \phi| |\phi(1 - \phi)| \quad (8)$$

3. Numerical simulation and analysis

3.1. Simulation model and boundary condition setting

A simplified 2D model of the needle-type dispenser was established by AUTOCAD, with the model geometry parameters are exhibited in figure 2(a), with point A in the figure for monitoring the pressure of the jetting chamber. In addition, the detailed boundary conditions of the model are shown in figure 2(b), and the initial working parameters are given in table 1. Subsequently, the model was then imported into the commercially available software package COMSOL Multiphysics 6.0, where the reciprocal motion of the needle was

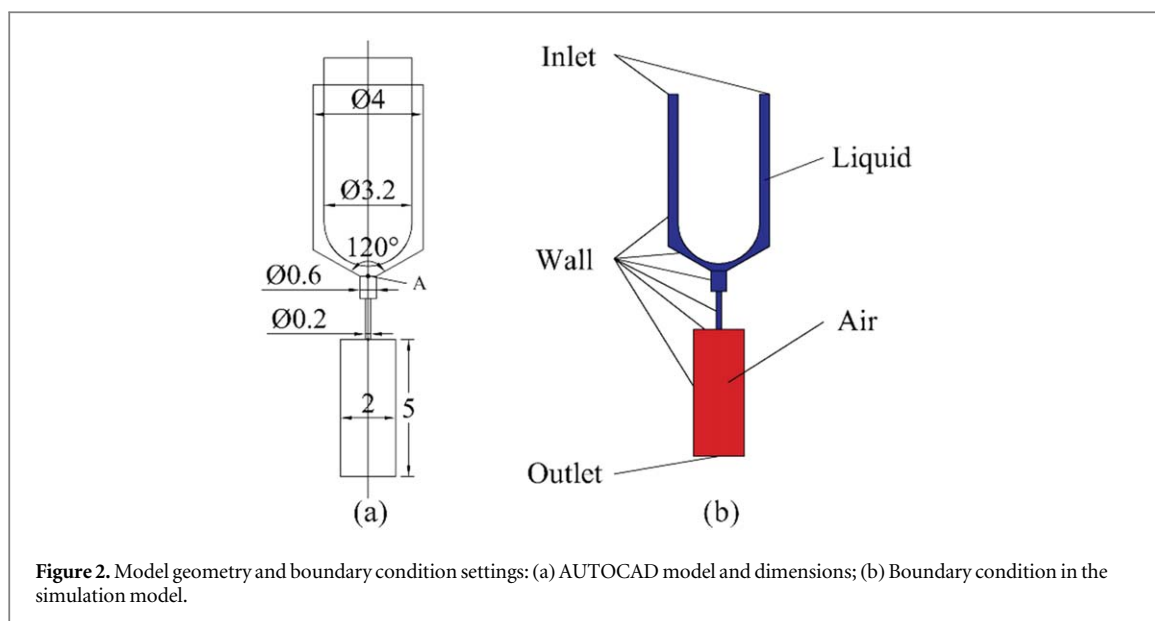


Figure 2. Model geometry and boundary condition settings: (a) AUTOCAD model and dimensions; (b) Boundary condition in the simulation model.

Table 1. The initial operating parameters in the simulation model.

Operating parameters	Values
Inlet pressure (MPa)	0.4
Outlet pressure (MPa)	0
Needle displacement (mm)	0.25
Fluid density (kg m^{-3})	1260
Fluid viscosity (Pa·s)	1
Surface tension (N m^{-1})	$7\text{e-}2$

commanded through the dynamic grid interface of COMSOL. Ultimately, the two-phase flow-level set module and the dynamic grid interface were used to couple the solution. Given the consideration of the strong coupling effect between the moving grid and the two-phase flow physical field, the direct coupling solver is likely to induce memory overflow. Hence, a segregated solver was utilized here, and the maximum number of iterations was modified to $5\text{e}3$ with a relative tolerance of $5\text{e-}4$. The time step was placed at $1\text{e-}6$ s.

3.2. Modeling methodology validation

In an effort to check the accuracy of the numerical scheme used in this manuscript. The corresponding simulation model was built, and the same boundary conditions were set according to [40]. The simulation outcome based on the R-LSM is compared with [40] as schematically illustrated in figure 3. As can be seen from figure 3 that the simulation results are in good agreement with the research results in [40]. It is evidenced that the numerical method utilized in this article is reliable for simulating micro-drop production.

3.3. Grid independence analysis

The grid resolution immediately determines the numerical computation precision. Generally speaking, the finer the grid division, the closer the numerical solution is to the real solution, though it will consume more computational resources. What is more, when the meshing is adequately fine, the sensitivity of the numerical outcome of the variation in the mesh dimension is much decreased. In this paper, we mesh the geometric modeling using free triangles to verify the grid independence by implementing the doubling and halving strategies for the number of meshes as proposed by Zhong *et al* [30]. The simulation results for various grid quantities are shown in figure 4, and the detailed meshing and the effect on droplet volume are shown in table 2. It can be observed from figure 4 that the change of droplet volume is very small when the mesh quantity reaches a certain value. This demonstrates that the element size we employed can exactly characterize the micro-drop jetting procedure. To balance the calculating precision and computational overhead, we confirmed the maximum mesh size in numerical modeling to be 0.04 mm, the number of grids to be 26912, and the average mesh quality to be 0.9464.

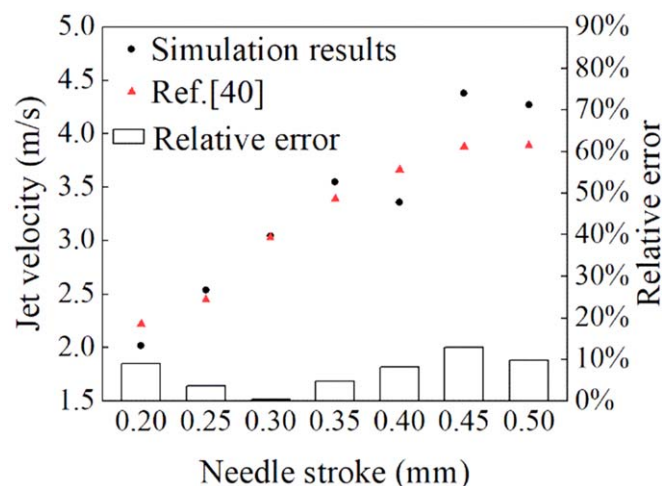


Figure 3. Comparison of the jet velocity for different needle strokes between simulation results and [40].

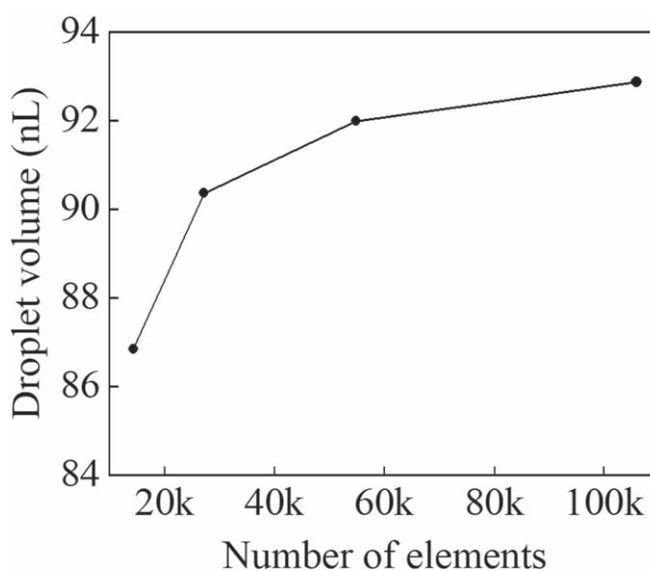


Figure 4. Grid independence analysis.

Table 2. Effect of mesh size on droplet volume.

Maximum grid size (mm)	Number of elements	Average mesh quality	Droplet volume (nL)
0.02	104486	0.9601	93.04
0.028	54322	0.9527	92.17
0.04	26912	0.9464	91.14
0.055	14179	0.9381	87.06

3.4. Mass conservation verification

To further improve the credibility of our simulation results, we verified the mass conservation of the LSM and explored the effect of grid size on it. Figure 5 illustrates the variation of the total mass flow rate of the system during 0–600 microseconds and the effect of grid size on the mass conservation of the LSM. It can be observed from figure 5 that the system is a non-stationary process in the 0–300 microseconds time period and the mass flow rate changes somewhat with time. The reason may be that when the fluid in the system starts to flow, it cannot flow out of the outlet in time due to the obstruction of the strong surface tension and viscous force of the adhesive, and at this moment the fluid in the working chamber will be compressed, which leads to a slight

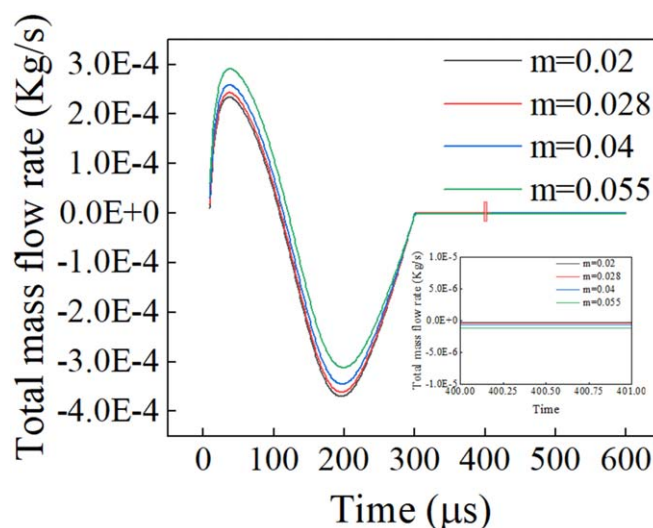


Figure 5. The verification of mass conservation and analysis of the effect of mesh size.

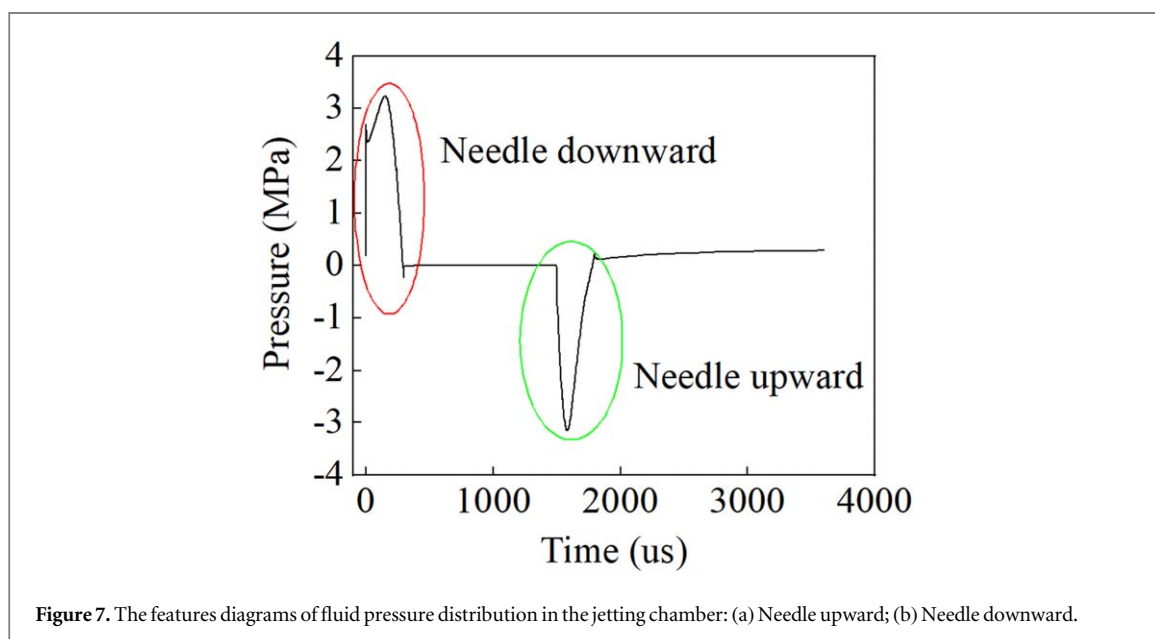
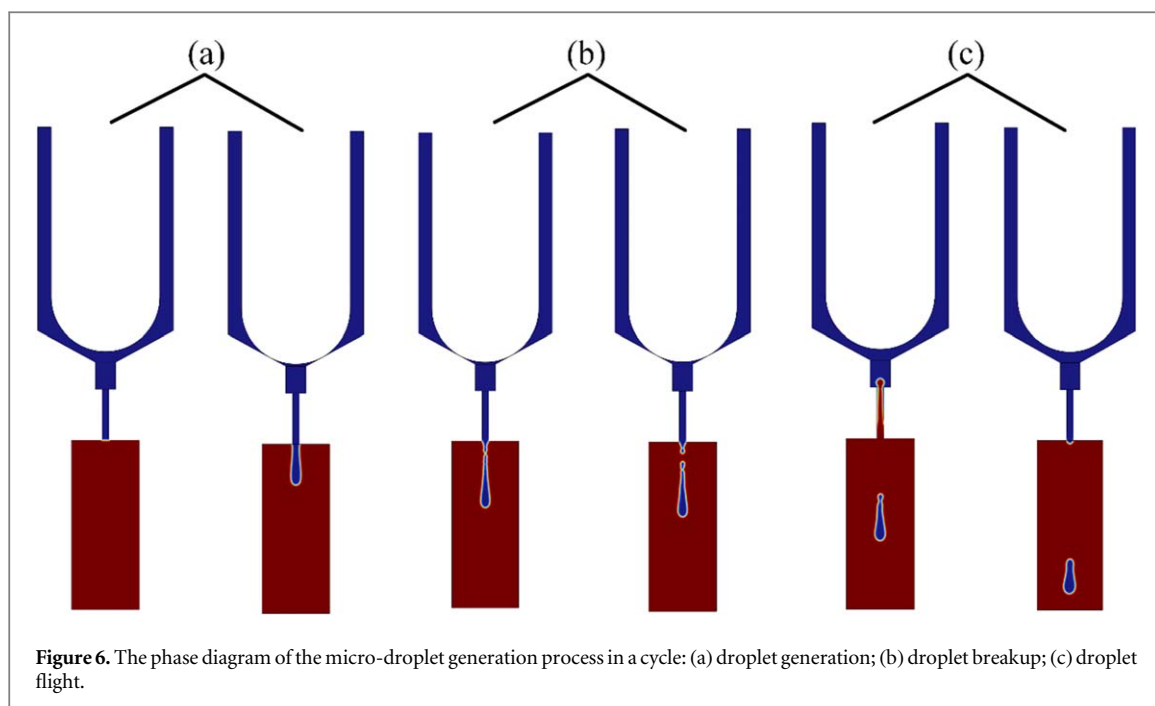
increase in the total mass flow rate of the system. As time progresses further, the gap between the needle and the nozzle decreases until it reaches to zero, and the liquid in the nozzle chamber is subjected to large velocity fluctuations due to transient pressure, which results in the mass flow rate at the outlet being greater than that at the inlet, i.e., there is a loss in the total mass flow rate. It should be noted that this range of mass fluctuations is acceptable, because the system total mass flow rate is always less than the relative tolerance of the solver (the relative tolerance is $5e-4$). And the system mass flow rate does not change with time in the time period of 300–600 microseconds, Its total mass flow rate (About $1e-6$ orders of magnitude) is much smaller than the solver tolerance which can be demonstrated that the mass of the LSM is kept constant in the quasi-static state.

Furthermore, we also analyze the influence of the grid size on the mass conservation of the LSM, and in figure 5, we can see that under dynamic conditions, the change in the total mass rate of the system can be more clearly captured by using a finer grid; under static conditions, the total mass flow rate of the system is infinitely close to 0 as the fineness of the grid increases, i.e., there is no significant mass loss in the system.

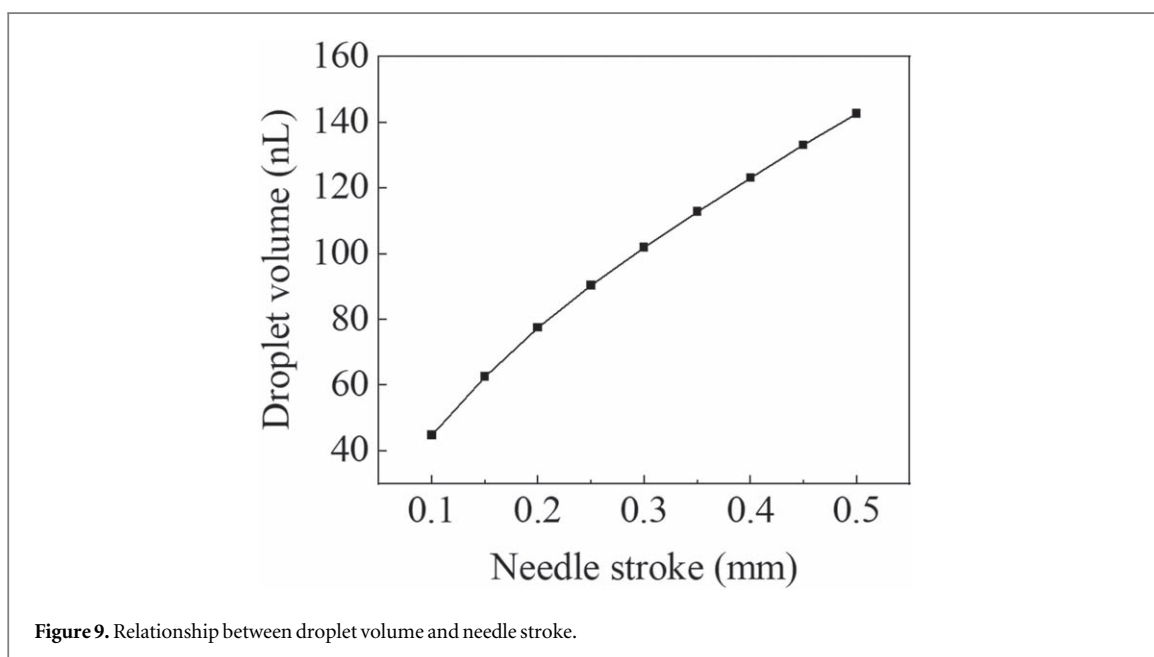
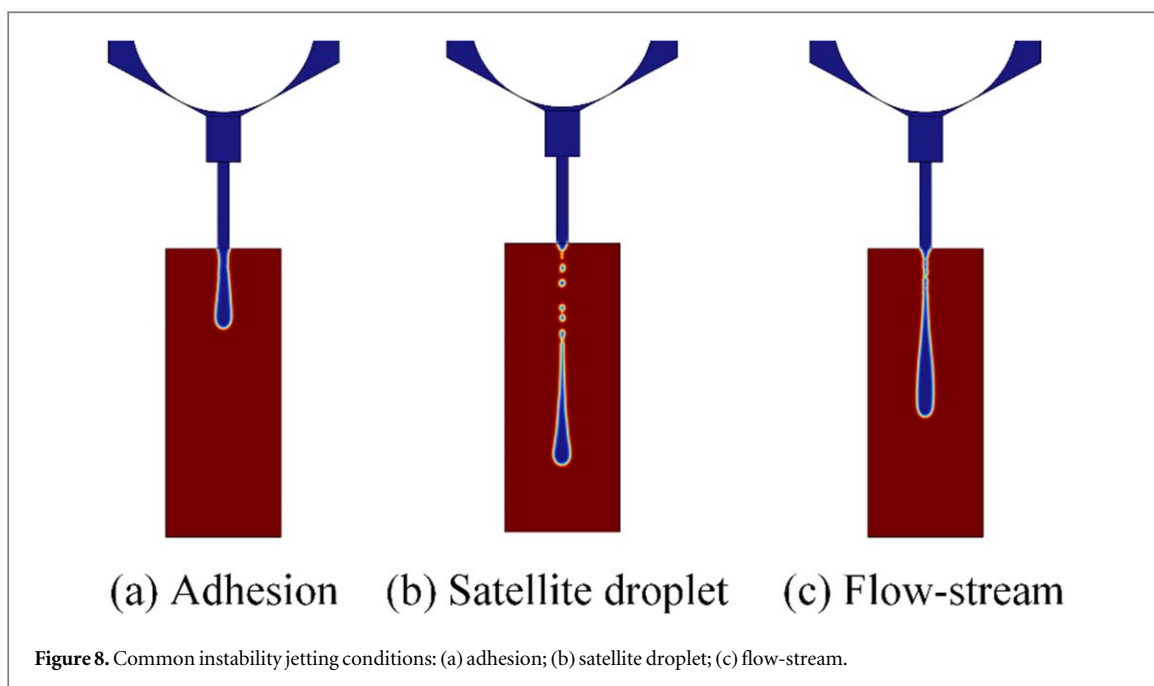
3.5. The process of droplet generation

Taking into account the resolution issue of moving mesh in the numerical simulation. The needle is placed at the maximum displacement at the initial moment, and the adhesive liquid fills the working chamber. The phase diagram of the micro-drop formation process in one period is illustrated in figure 6, where the blue area represents the adhesive material. In this figure, the micro-droplet generation procedure is divided into three phases: droplet generation, droplet breakup, and droplet flight. In the first stage (see figure 6(a)), the adhesive fluid in the working cavity is squeezed by an external force that makes the liquid flow out of the nozzle. The outward force here comes from two sources: One is provided by the feed pressure, and the other is the force generated by the movement of the needle. Besides, the adhesive in the working chamber appears in two various flow states: most of the adhesive flows into the nozzle, while the remaining part flows backward along the space of both the working cavity and the needle. In the second stage (see figure 6(b)), the working chamber and the nozzle will no longer be linked when the needle moves to the lowest point. At this point, the jet chamber will produce a local transient high pressure, which will increase the adhesive's flow rate. Under the effect of surface energy and kinetic energy, the tail of the adhesive will be elongated until it breaks into droplets. In the final stage (see figure 6(c)), the needle will move upward to its original position. During this process, the pressure value in the injection chamber will change to negative because the motion of the needle is much faster than the flow velocity of the fluid. This causes a small amount of air and residual adhesive fluid to be sucked back into the nozzle while the droplets keep on moving toward the substrate under their own gravity and kinetic energy. Simultaneously, the adhesive in the supply reservoir will refill the working cavity.

The level of pressure in the injection chamber determines the droplet injection speed and also affects the droplet forming quality. Therefore, we performed a detailed analysis of the pressure variation in the injection chamber during the micro-drop injection. As can be seen from figure 7, when the needle moves downward, the pressure in the injection chamber shows a trend of increase - slight decrease - slow increase - large decline. The reason for this phenomenon is that the speed of the needle does not start from zero in the simulation but has a certain initial rate. Moreover, as the needle movement disturbs the equilibrium of the flow field, the liquid in the



working chamber is extruded and begins to flow, thus contributing to a small drop in the pressure of the ejection chamber. After that, as the interval between the needle and the nozzle decreases continuously, the pressure in the jetting cavity gradually increases causing the fluid to flow out of the nozzle. However, owing to the viscous shear force and surface tension, the liquid does not disengage directly from the nozzle, instead, an accumulation is built up at the nozzle orifice. When the needle hits the nozzle, the pressure in the jetting chamber will reach a maximum value of 3.4 MPa. At the current time, because the supplementary adhesive cannot flow in time, a short vacuum will be formed in the injection chamber, so the pressure in the injection chamber will change from the highest value to a negative value instead of directly to 0. When the needle is in an upward movement, the pressure of the injection chamber shows a downward-upward trend. This is because the speed of needle motion is much greater than the fluid flow rate; the jet chamber will form a certain negative pressure. At the same time, the feed fluid will soon fill the gap generated by the needle motion, so the pressure of the injection chamber will appear to drop and then rise. From the above analysis, it can be seen that the needle motion is the primary reason for the change of stress in the jetting cavity.



4. The influence of coupling parameters

Micro-fluid injection is a complicated multi-physical field coupling process, its formation process is affected by lots of coupling parameters, such as surface tension, needle stroke, nozzle diameter, supply pressure, and fluid viscosity. If the relevant parameters are not set properly, it will cause adhesion, satellite droplets and flow-stream unstable jet behaviors as shown in figure 8. Next, we assess the effect of the relevant coupling parameters on the droplet volume and explored the unstable jet behaviors that may be induced.

4.1. Needle stroke

In an effort to examine the influence of needle stroke on droplet ejection volume, the needle strokes were set to 0.1–0.5 mm, the step length was kept to 0.05 mm, and the rest of the parameters are presented in 3.1. Figure 9 exhibits the relationship between needle stroke and droplet volume, from which it can be seen that much fluid flows out of the nozzle as the needle stroke gains. This is explained by the fact that the motion time of the needle is constant. In other words, the larger the needle stroke, the greater the velocity of fluid motion will increase, and

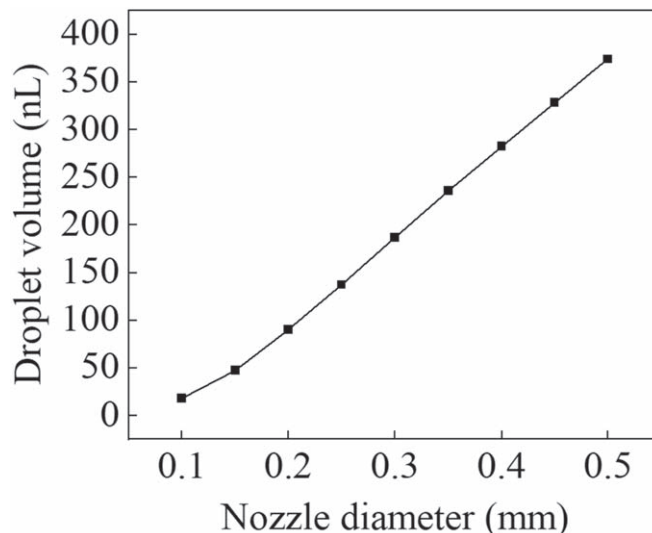


Figure 10. Relationship between droplet volume and nozzle diameter.

thus the droplet volume will increase. Furthermore, we also found in the simulation that when the needle stroke was less than 0.2 mm, a large amount of adhesive fluid would hang on the nozzle instead of forming a single droplet, probably because the momentum of the fluid was not enough to overcome the surface tension and viscous force of the fluid. When needle travel is greater than 0.3 mm, the fluid acquires a large acceleration, which causes the main droplet to carry one or more satellite droplets behind it and eventually evolve into a flow-stream. Therefore, in the real dispensing process, when additional parameters maintain unchanged, the needle stroke should be regulated within a defined range to evade abnormal jetting.

4.2. Nozzle diameter

The nozzle diameter is an essential element affecting the droplet jetting volume. In order to investigate the effect of nozzle diameter on droplet volume, the nozzle diameter was varied in the range of 0.1 to 0.5 mm with an interval of 0.05 mm and all other factors being equal. From figure 10, as can be observed in the droplet volume increases with increasing nozzle diameter. When the nozzle diameter is over 0.35 mm, its volume is still increasing tendency, although the ratio of change has decreased. This reason is that the fluid velocity is much smaller than the critical jet velocity, causing the fluid to be unable to be effectively injected and accumulate at the nozzle. When the nozzle diameter is below 0.15 mm, the variety in drop volume is smaller. This is due to the viscous effect of the liquid that enlarges with the diminution of nozzle diameter, and the fluid will hang on the nozzle orifice at this moment. Consequently, in engineering applications, the nozzle should be designed according to the specific application requirements of the appropriate size.

4.3. Supply pressure

In the period of micro-fluid injection, the prominent role of the supply pressure is to provide a specific driving force for the liquid flow in the working chamber. In order to evaluate the effect of the supply pressure on the injection volume, the supply pressure was enhanced from 0.1 MPa to 1 MPa, and it is manifest from figure 11 that the simulation results are under diverse supply pressures. The consequences prove that the supply pressure is approximately linearly proportional to the droplet volume when other simulation conditions remain unchanged. That is, the larger the supply pressure, the larger the droplet volume. The reason for this that as the supply pressure enhances, the liquid flow rate in the working chamber is speeded up, so the drop volume increases. Else, we also found that a larger supply pressure is beneficial to shorten the drop formation time, but when the supply pressure is beyond 1 MPa, there is satellite droplet generation. It is important to emphasize that the higher the supply pressure, the higher the requirement for the sealing property of the injector. Hence, a rational supply pressure should be placed to avoid erratic jetting or leakage of fluid.

4.4. Fluid viscosity

Viscosity is an important indicator of fluid flow. The greater the viscosity, the worse the liquid flow will be, as well as prolonging the time for droplet formation and separation and requiring more energy to accomplish injection. As for this section, we investigate the effect of fluid viscosity on droplet volume. The viscosity was placed at varying in the 0.2–2 Pa·s range with a step size of 0.2 Pa·s, and other parameters were not changed. The

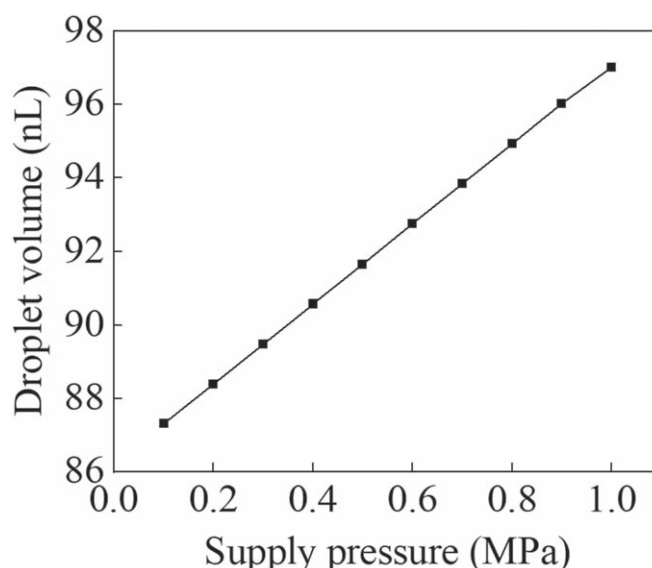


Figure 11. Relationship between droplet volume and supply pressure.

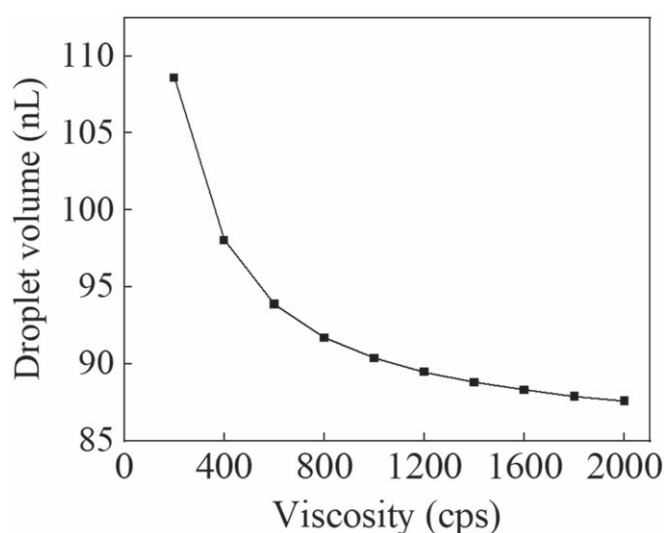


Figure 12. Relationship between droplet volume and fluid viscosity.

droplet volume versus fluid viscosity is illustrated in figure 12, which shows that the droplet volume decreases as the viscosity of the fluid increases. When the liquid viscosity exceeds 1.6 Pa·s, the variation in droplet volume is comparatively slow. This happens because as the adhesive fluid viscosity adds, the viscous effect significantly enhances, and the droplet cannot separate from the nozzle in a short time, thus making its spraying time longer. As the viscosity of the adhesive liquid is lower than 0.8 Pa·s, the drop volume shows an opposite tendency with the decline of viscosity. The fundamental causation for this is that the viscous resistance of the liquid is enormously reduced; at this moment, the liquid is dominantly ejected from the nozzle in the form of satellite droplets or flow-stream.

4.5. Surface tension

Surface tension has an important effect not only on the shape of a droplet, but also on the droplet volume. In order to investigate the effect of surface tension on droplet volume, the surface tension was set to 0, 1e-2, 3e-2, 5e-2, and 7e-2 N m⁻¹ and other parameters were kept constant. From figure 13, it can be seen that the droplet volume increases proportionally with the increment of surface tension. In addition, we also found that the slope of the enhancement of droplet volume increases significantly when the surface tension is greater than 3e-2 N m⁻¹, which may be due to the fact that the larger surface tension induces the accumulation of more liquid,

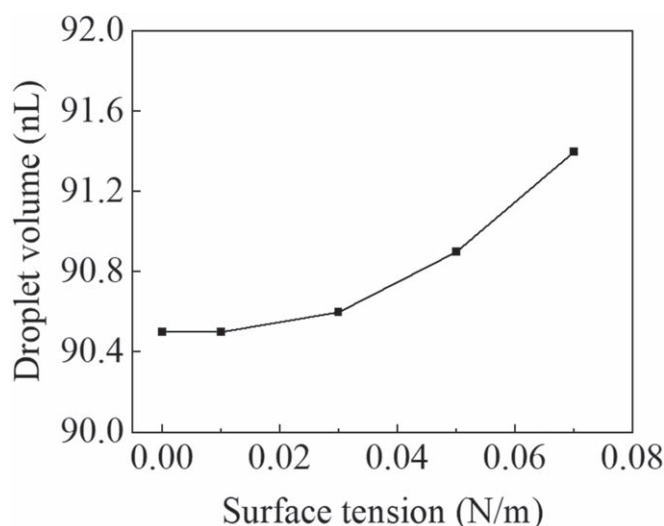


Figure 13. Relationship between droplet volume and surface tension.

resulting in the formation of larger droplets. Eventually, it should be noted that a larger surface tension increases the difficulty of droplet ejection.

5. Summary and conclusions

This article introduced a needle-type injector that can realize high-viscosity droplet generation. A LSM for investigating the mechanism of highly viscous droplet formation was presented. Furthermore, the influence law of system coupling parameters on droplet formation volume was analyzed. The main conclusions obtained are as follows:

- (1) The process of high-viscosity droplet generation can be roughly classified into three stages, i.e., droplet formation, separation, and droplet flight. In addition, the reasons for the pressure variation in the injection chamber were discussed in detail, and the maximum injection pressure of 3.4 MPa was obtained for this injector.
- (2) The effects of needle displacement, nozzle diameter, supply pressure, fluid viscosity, and surface tension on droplet volume were analyzed. We obtained that the nozzle diameter has the greatest effect on the droplet volume, and the surface tension has the minor effect. In addition, we found that increasing the needle displacement, increasing the nozzle diameter, increasing the supply pressure, increasing the surface tension, or reducing the fluid viscosity is beneficial to droplet formation. However, once a certain threshold is exceeded, unstable injection phenomena such as adhesion, satellite droplets, and jets are caused.

Overall, this study intensifies the understanding of the process of high-viscosity droplet generation in a needle-type injector, thus providing some reference to the stable generation of high-viscosity droplets.

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Data availability statement

All data that support the findings of this study are included within the article (and any supplementary files).

Compliance with ethical standards

Conflict of interest

The authors declare no conflict of interest

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